

Lab 3-2: Transformers

<삼성 AI 전문가 교육과정> 실습
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2025.05.29

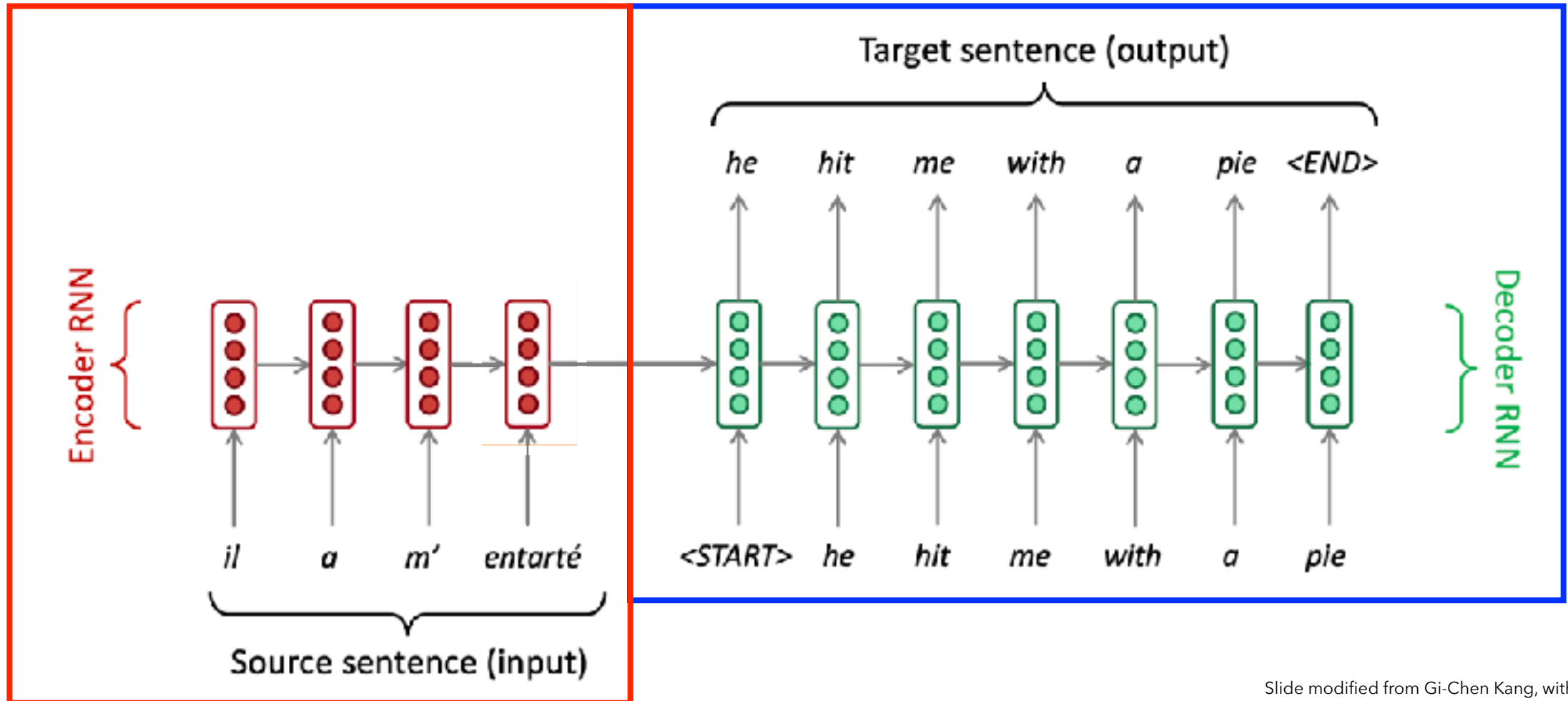
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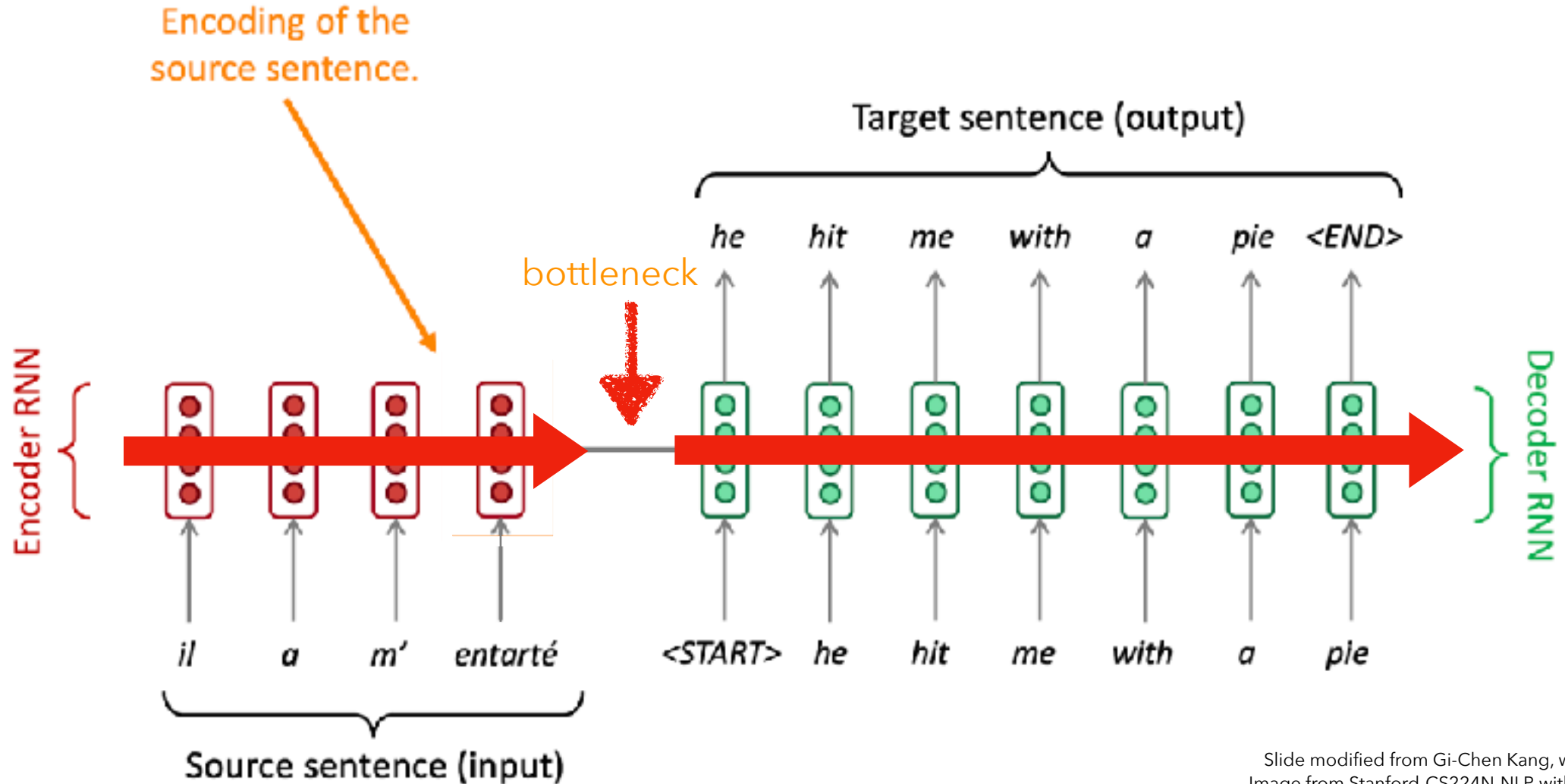
- Sequence-to-Sequence Model
- Attention Mechanism
- Transformer
- BERT
- GPT

Sequence-to-Sequence Models



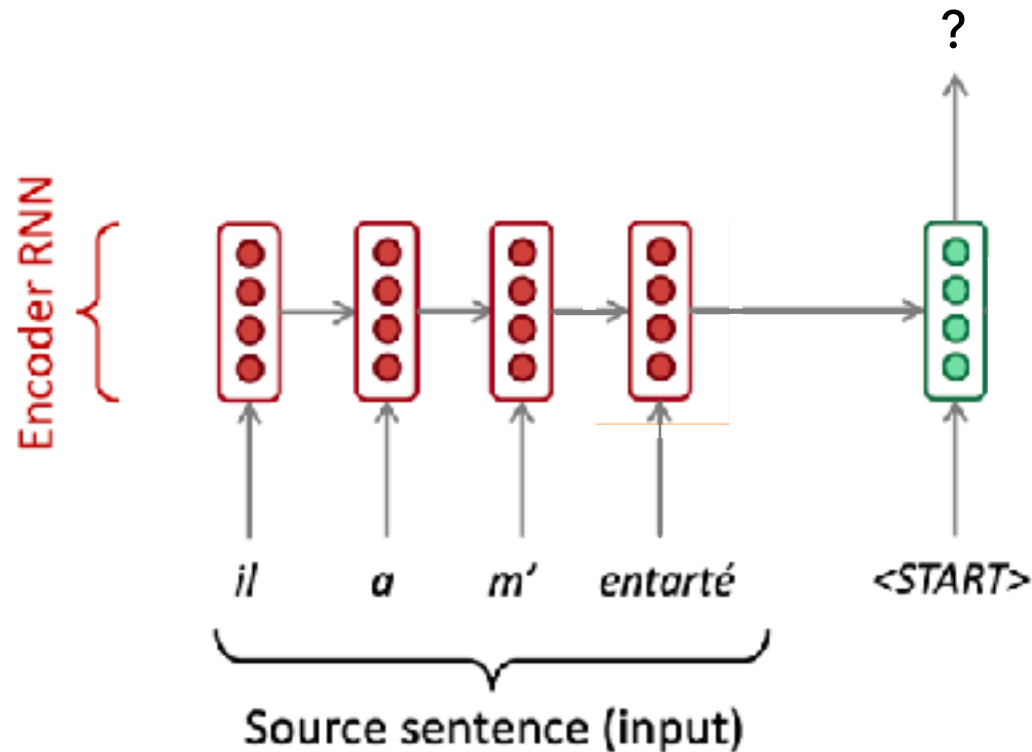
The Limitations of Seq2Seq w/ RNN

- RNN encoder compresses source sentence into a single vector -> information bottleneck



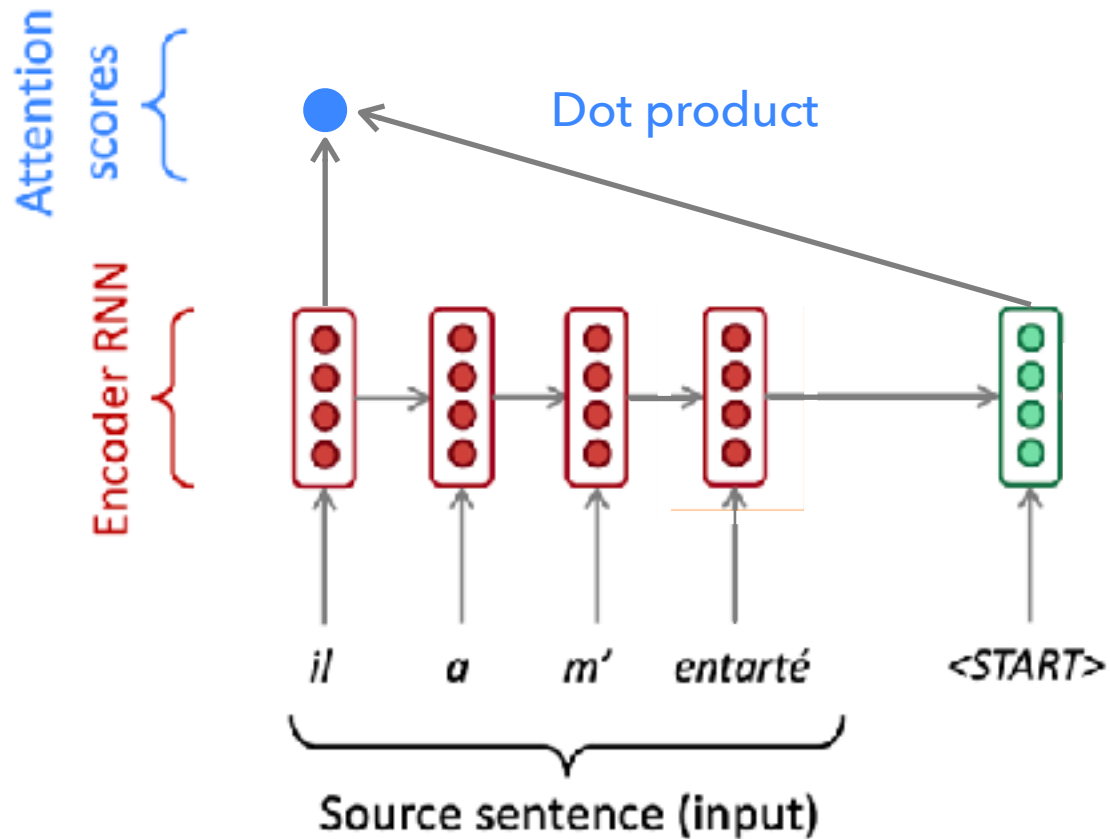
Sequence-to-Sequence with "Attention"

How to address the bottleneck problem?



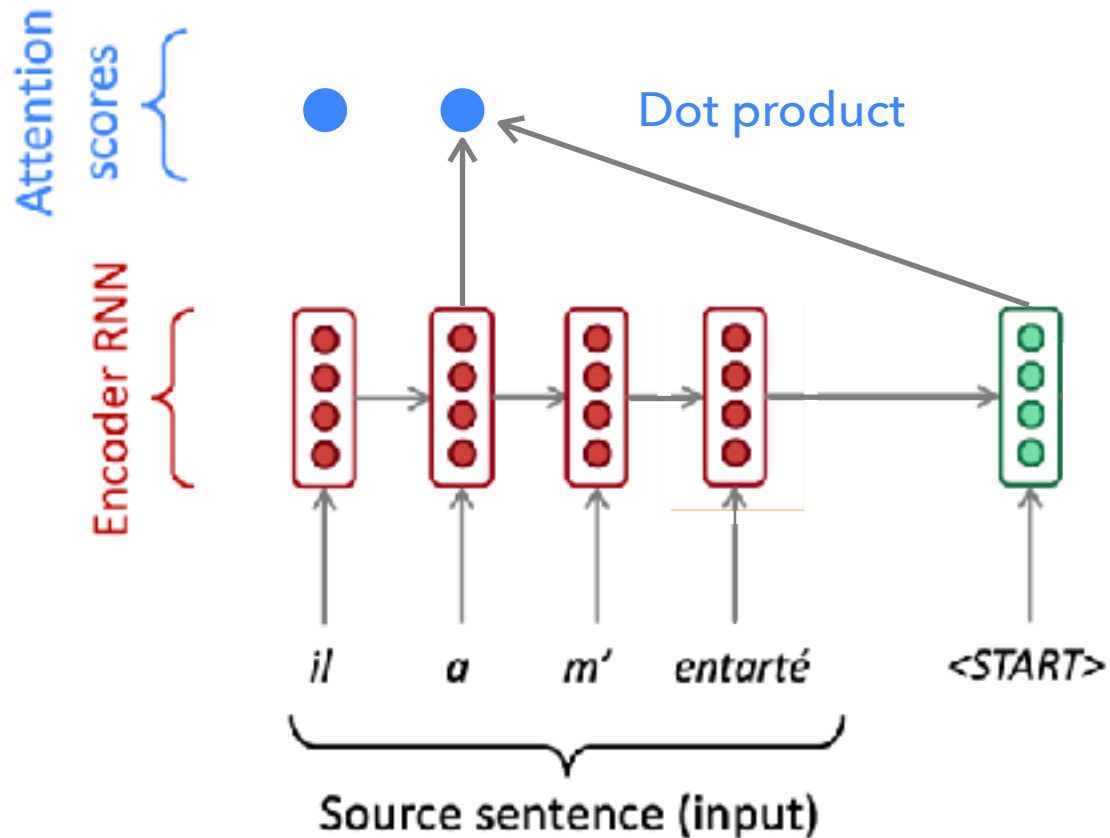
Sequence-to-Sequence with "Attention"

Connection to the each step of the encoder!



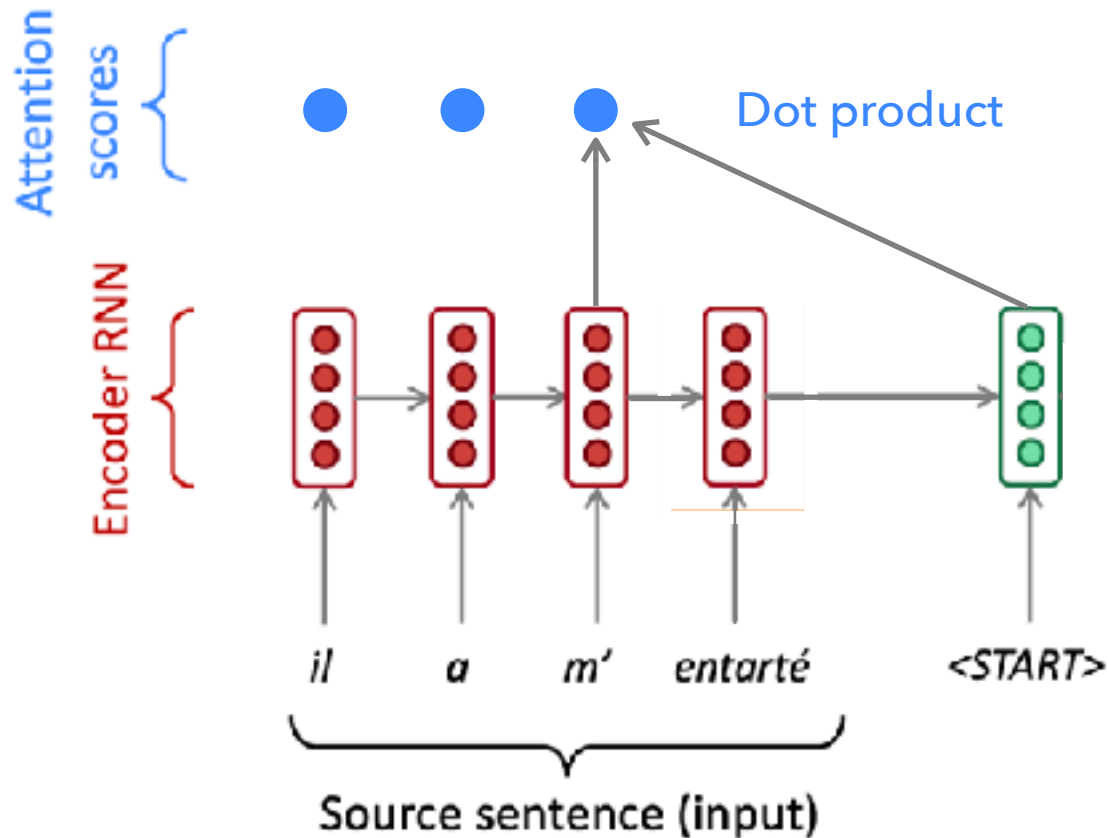
Sequence-to-Sequence with "Attention"

Connection to the each step of the encoder!



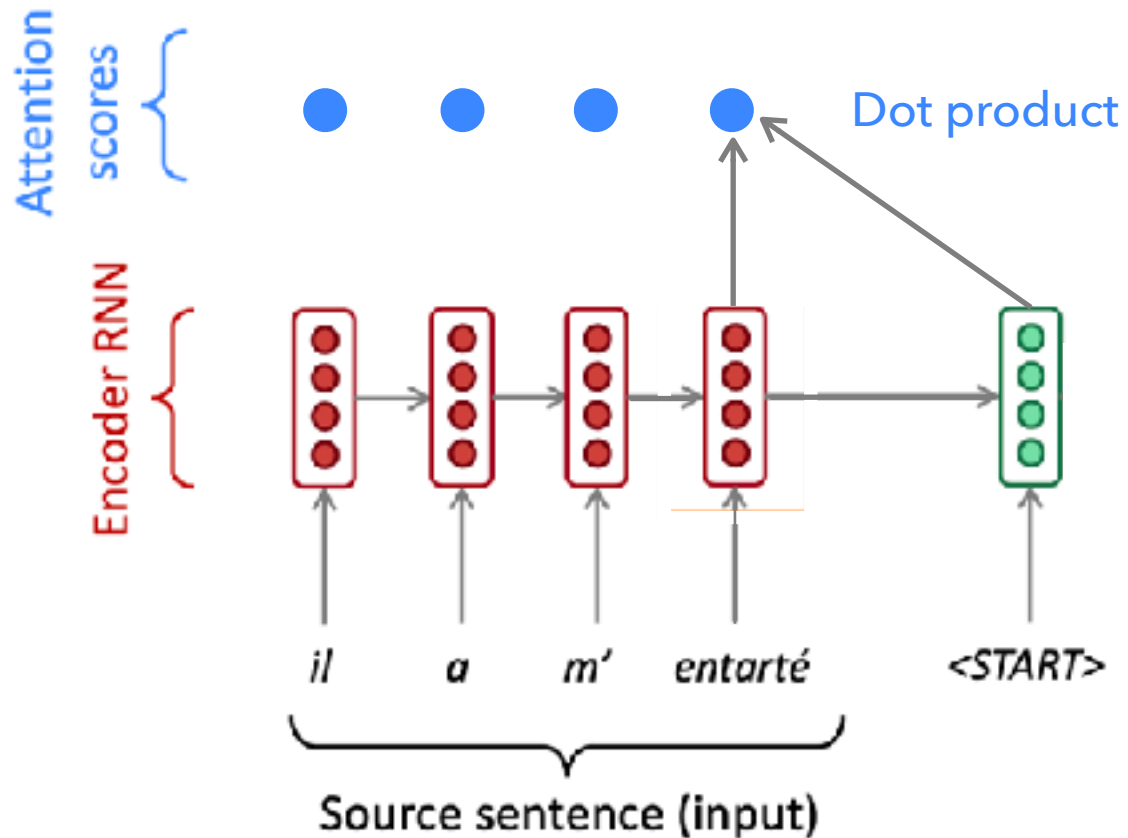
Sequence-to-Sequence with "Attention"

Connection to the each step of the encoder!

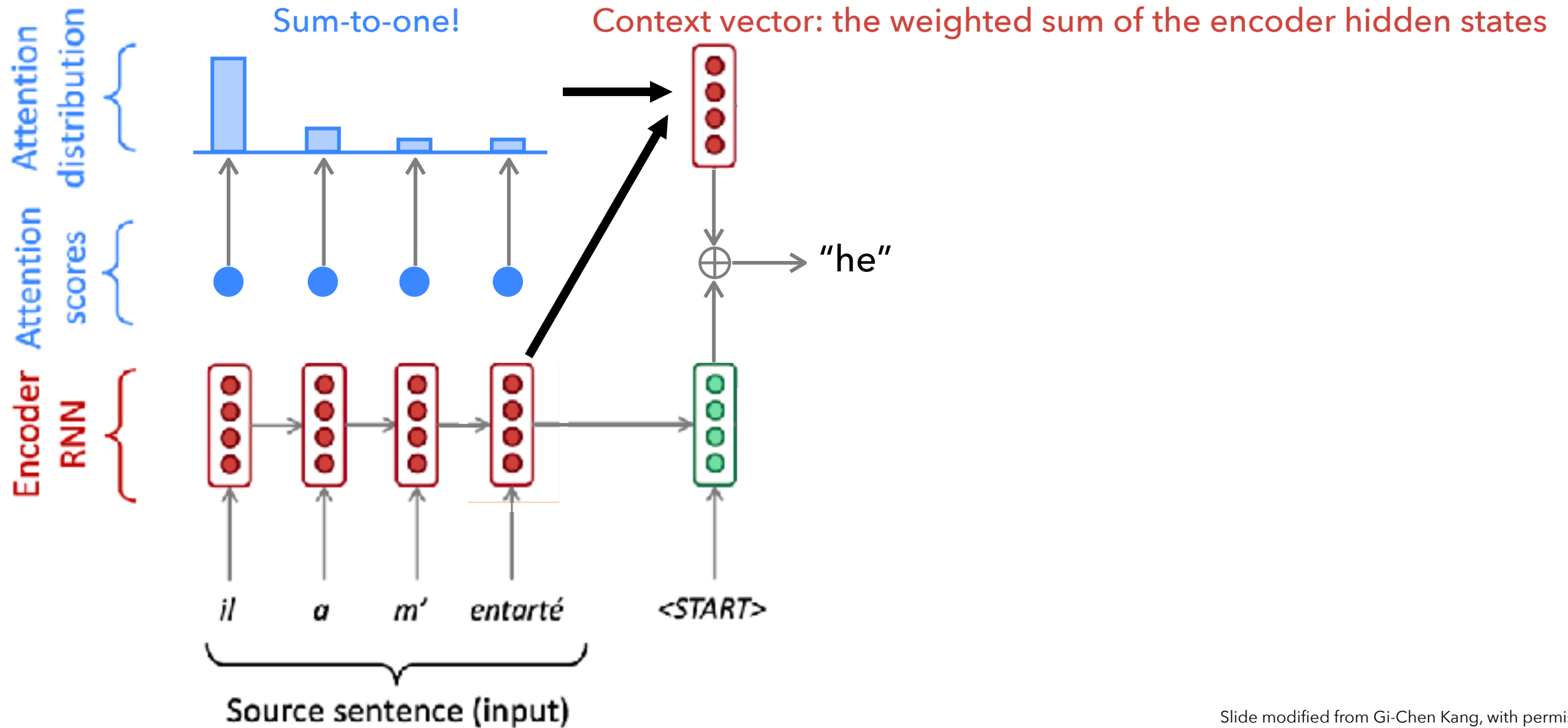


Sequence-to-Sequence with "Attention"

Connection to the each step of the encoder!

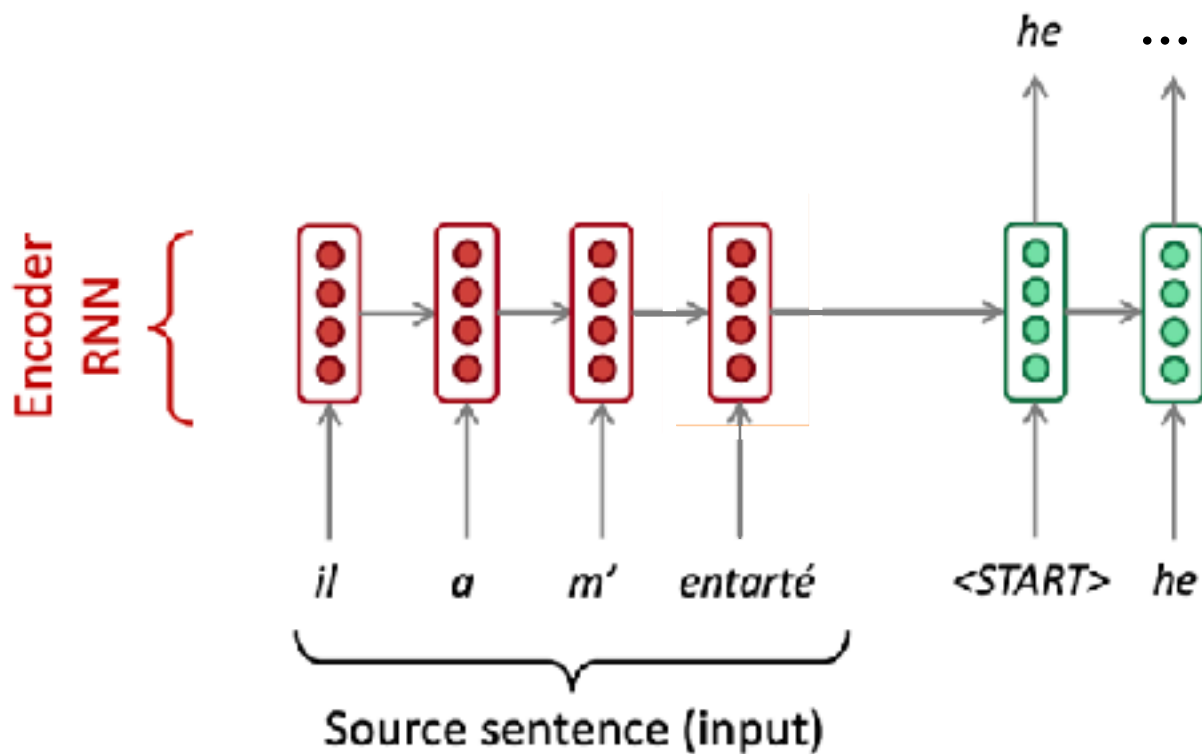


Sequence-to-Sequence with "Attention"

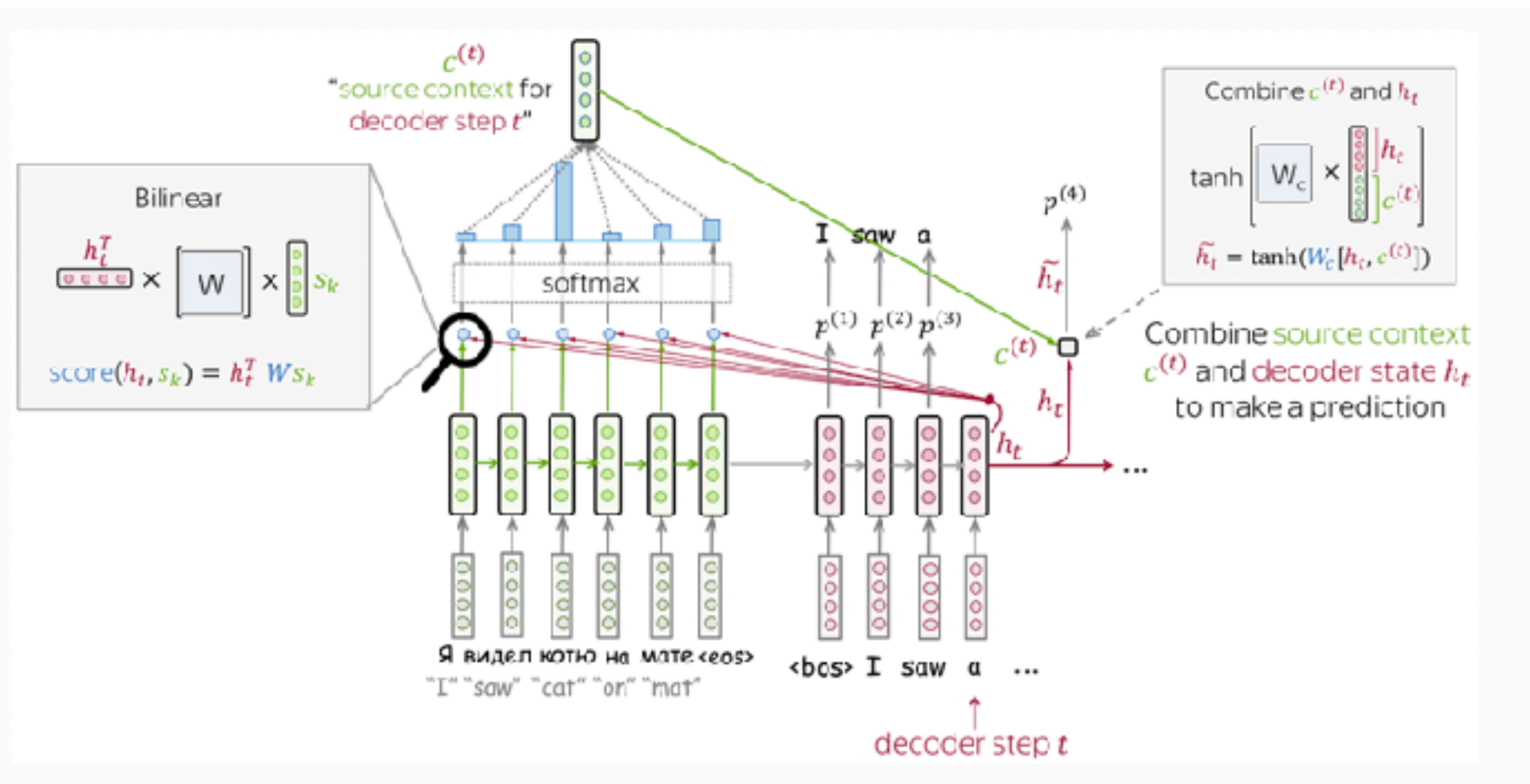


Sequence-to-Sequence with "Attention"

Repeating the attention mechanism for every step

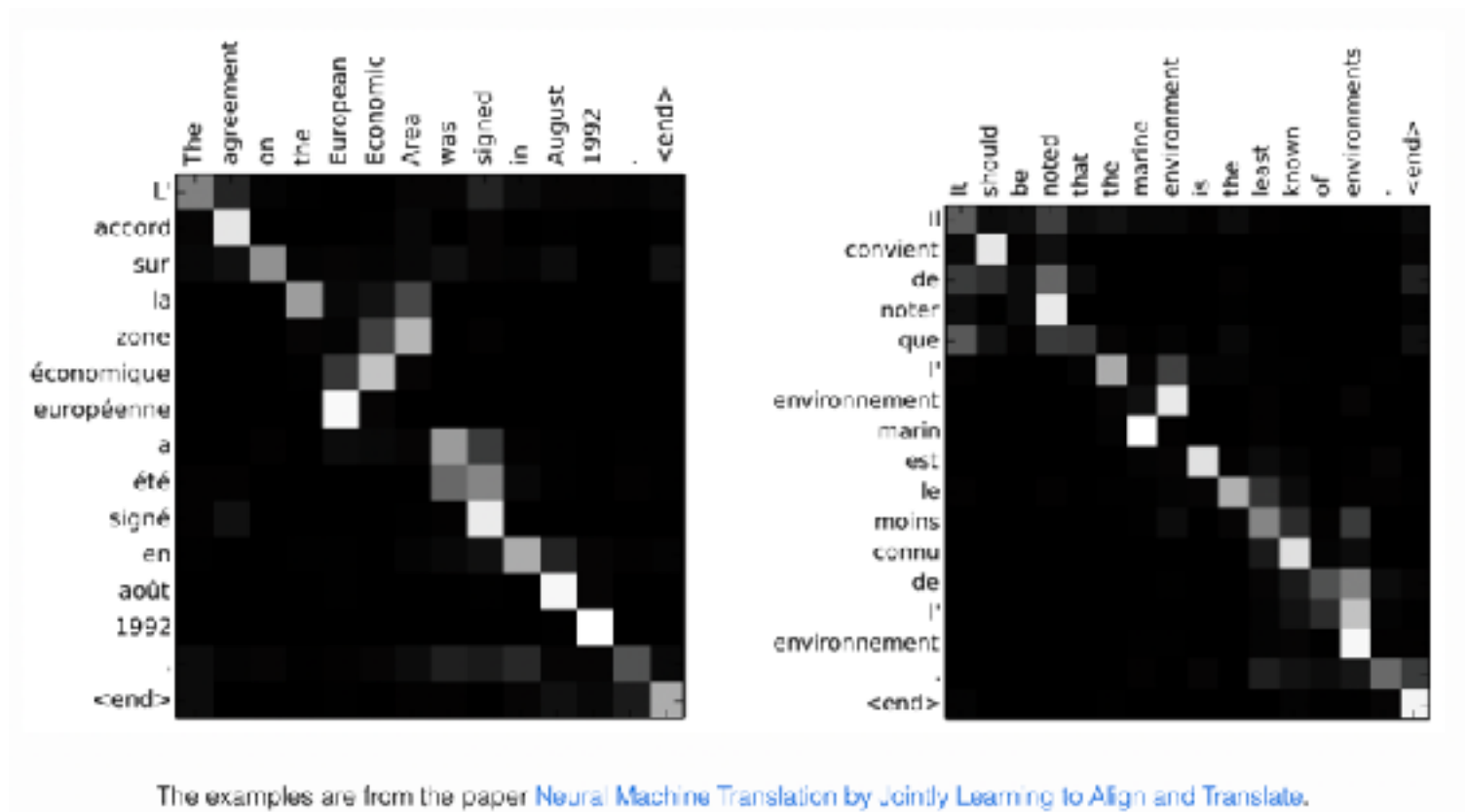


Seq2Seq + Attention



Seq2Seq + Attention

Attention learns Alignment



Source :

reference : [Neural Machine Translation by Jointly Learning to Align and Translate](#) ICLR2015

Transformer

Attention is all you need NIPS 2017

Attention Is All You Need

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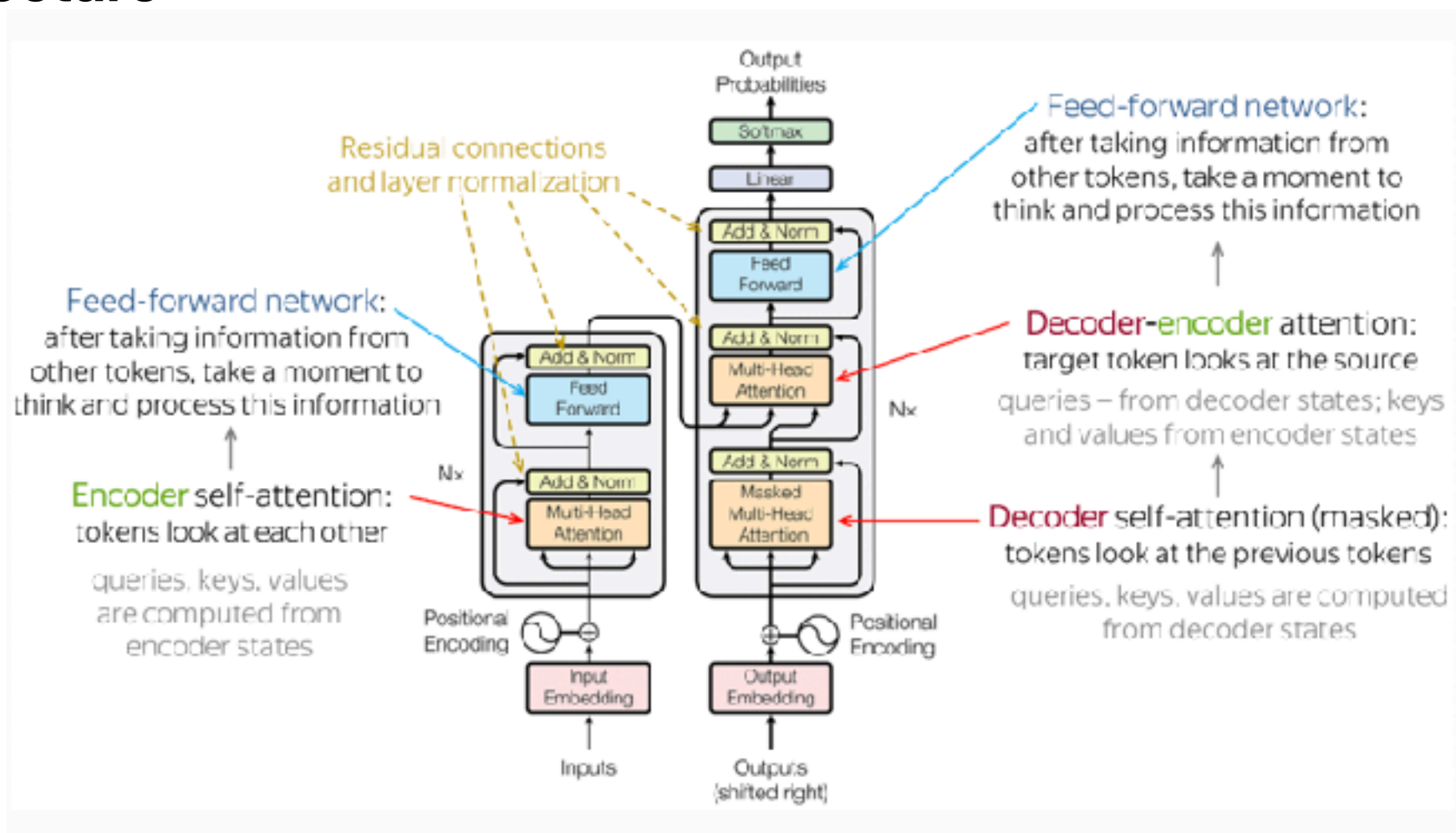
Transformer

Only attention !

	Seq2seq without attention	Seq2seq with attention	Transformer
processing within encoder	RNN/CNN	RNN/CNN	attention
processing within decoder	RNN/CNN	RNN/CNN	attention
decoder-encoder interaction	static fixed- sized vector	attention	attention

Transformer

Architecture



Transformer

Attention

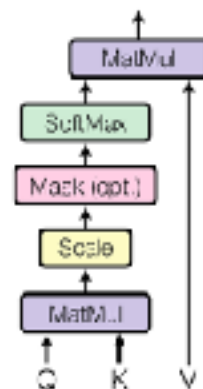
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$

Where the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and $W^O \in \mathbb{R}^{hd_k \times d_{\text{model}}}$.

Scaled Dot-Product Attention



Multi-Head Attention

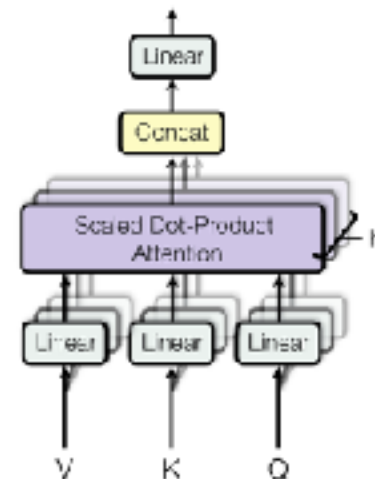


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

- Attention block은 아래의 두 가지 computation으로 해석할 수 있음
 - search(query-key interaction) : selection of a relevant entity
 - retrieval(value) : extraction of relevant features from the selected entity

Transformer

Self-Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad \begin{array}{l} Q = X \times W_Q \\ K = X \times W_K \\ V = X \times W_V \end{array}$$

Where the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and $W^O \in \mathbb{R}^{d_v \times d_{\text{model}}}$.

$$X = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$W_Q = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$W_K = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$W_V = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 3 & 0 \\ 1 & 0 & 3 \\ 1 & 1 & 0 \end{bmatrix}$$

$$Q = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 2 & 2 \\ 2 & 1 & 3 \end{bmatrix}$$

$$K = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 4 & 4 & 0 \\ 2 & 3 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 2 & 0 \\ 0 & 3 & 0 \\ 1 & 0 & 3 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 0 \\ 2 & 6 & 3 \end{bmatrix}$$

Transformer

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad \begin{array}{l} \mathbf{Q} = \mathbf{X} \times \mathbf{W}_Q \\ \mathbf{K} = \mathbf{X} \times \mathbf{W}_K \\ \mathbf{V} = \mathbf{X} \times \mathbf{W}_V \end{array}$$

Where the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

$$\mathbf{Q} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 2 & 2 \\ 2 & 1 & 3 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} 0 & 1 & 1 \\ 4 & 4 & 0 \\ 2 & 3 & 1 \end{bmatrix}$$

$$\sigma(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

$$\begin{bmatrix} 1 & 0 & 2 \end{bmatrix} \times \begin{bmatrix} 0 & 4 & 2 \\ 1 & 4 & 3 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 4 \end{bmatrix} \quad \rightarrow \quad \text{softmax}\left(\left[\frac{2}{\sqrt{3}}, \frac{4}{\sqrt{3}}, \frac{4}{\sqrt{3}}\right]\right) = [0.13613, 0.43194, 0.43194]$$

$$\begin{bmatrix} 0.13163 & 0.43194 & 0.43194 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 2 & 8 & 0 \\ 2 & 6 & 3 \end{bmatrix} = \begin{bmatrix} 1.8639 & 6.3194 & 1.7042 \end{bmatrix}$$

Transformer

Positional Encoding

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

- Sine and cosine functions of different frequencies
- We also experimented with using learned positional mbeddings [9] instead, and found that the two versions produced nearly identical results (see Table 3 row (E)).

Transformer

Architecture

Each vector receives three representations ("roles")

$$[W_Q] \times \begin{bmatrix} \text{green} \\ \text{green} \\ \text{green} \end{bmatrix} = \begin{bmatrix} \text{blue} \\ \text{blue} \\ \text{blue} \end{bmatrix}$$

Query: vector from which the attention is looking

"Hey there, do you have this information?"

$$[W_K] \times \begin{bmatrix} \text{green} \\ \text{green} \\ \text{green} \end{bmatrix} = \begin{bmatrix} \text{yellow} \\ \text{yellow} \\ \text{yellow} \end{bmatrix}$$

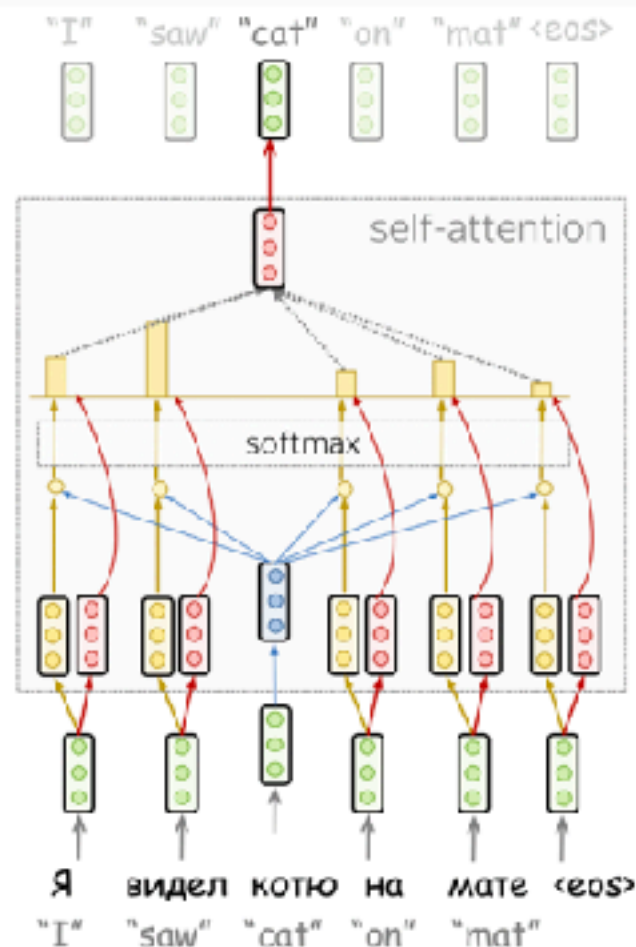
Key: vector at which the query looks to compute weights

"Hi, I have this information - give me a large weight!"

$$[W_V] \times \begin{bmatrix} \text{green} \\ \text{green} \\ \text{green} \end{bmatrix} = \begin{bmatrix} \text{red} \\ \text{red} \\ \text{red} \end{bmatrix}$$

Value: their weighted sum is attention output

"Here's the information I have!"



Transformer

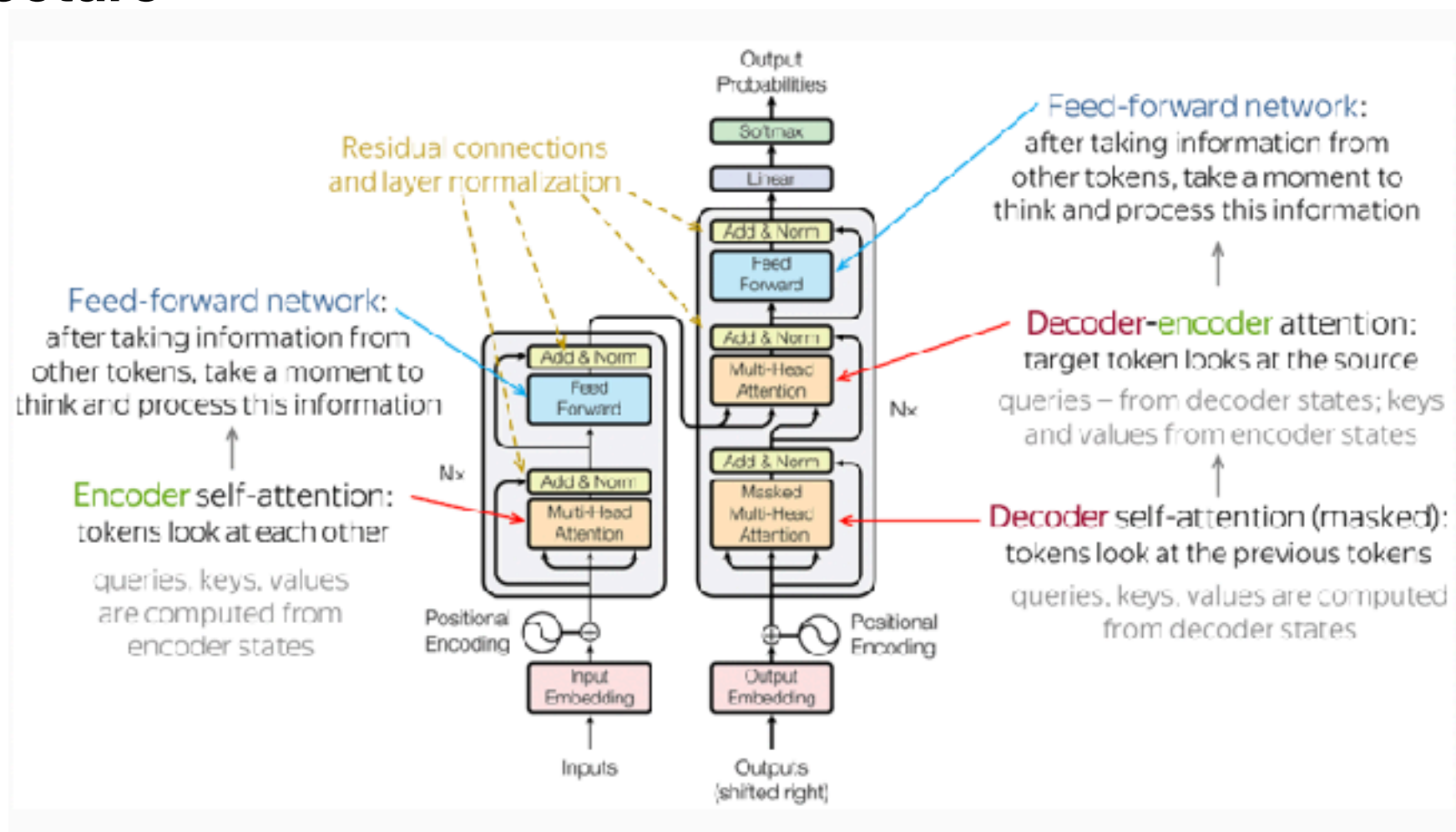
Architecture

Table 1: Maximum path lengths, per-layer complexity and minimum number of sequential operations for different layer types. n is the sequence length, d is the representation dimension, k is the kernel size of convolutions and r the size of the neighborhood in restricted self-attention.

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

Transformer

Architecture



Recent Language Models

■ BERT(Bidirectional Encoder Representations from Transformers)

- Masked Language Model
- 문장 중간에 빈칸을 만들고 해당 빈칸에 어떤 단어가 적절할지 맞춤
- 빈칸 앞뒤 문맥을 모두 살필 수 있음
 - 양방향(bidirectional)적

■ GPT(Generative Pre-trained Transformer)

- 전 단어들이 주어졌을 때 다음 단어가 무엇인지 맞춤
- 문장 시작부터 순차적으로 계산
 - 일방향(unidirectional)적

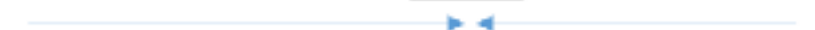
GPT

어제 카페 갔었어 거기 사람 많더라



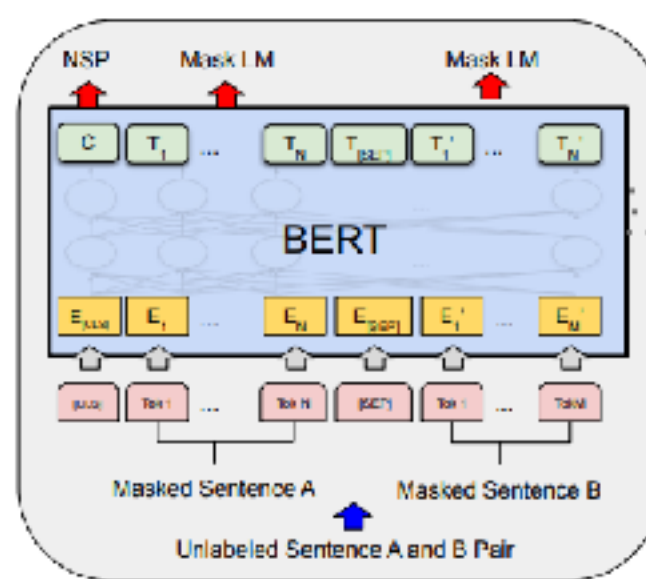
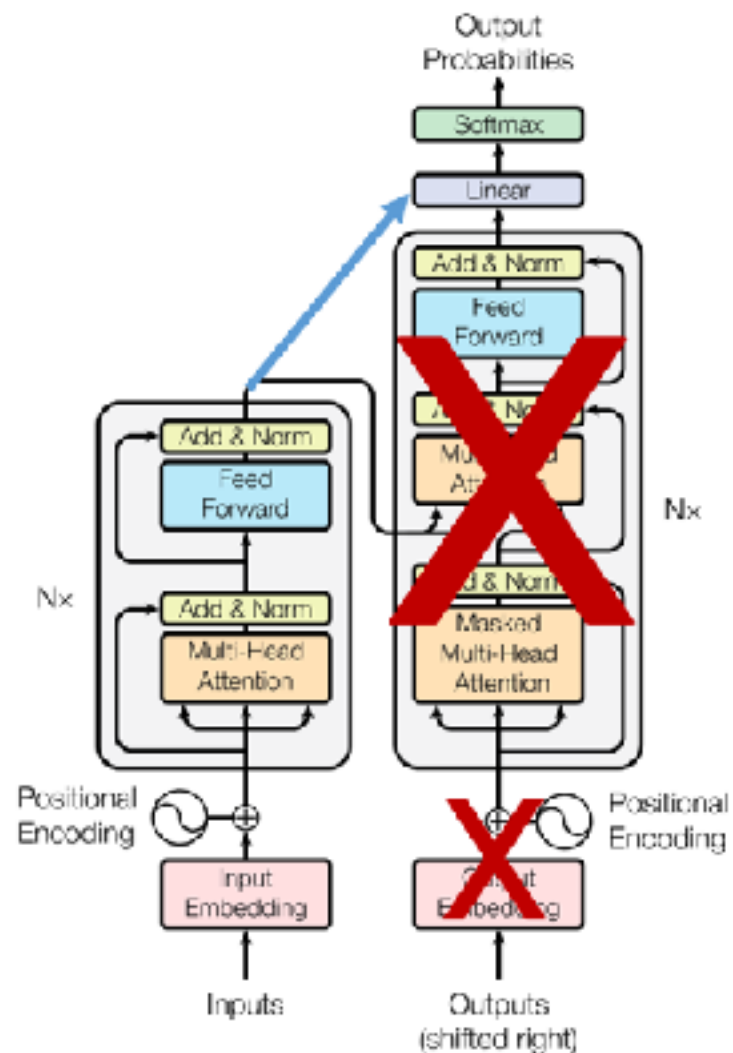
BERT

어제 카페 갔었어 사람 많더라

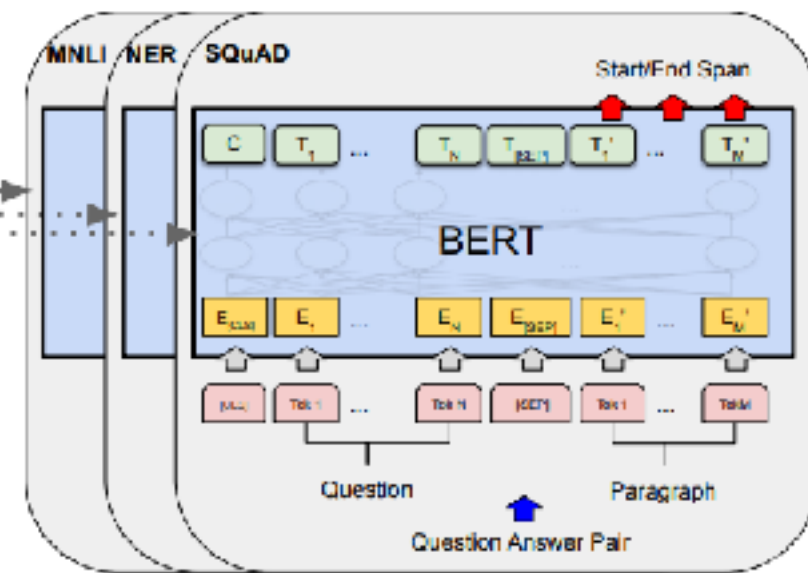


BERT

✓ only Encoder



Pre-training



Fine-Tuning

BERT – Training

■ Training with masked words

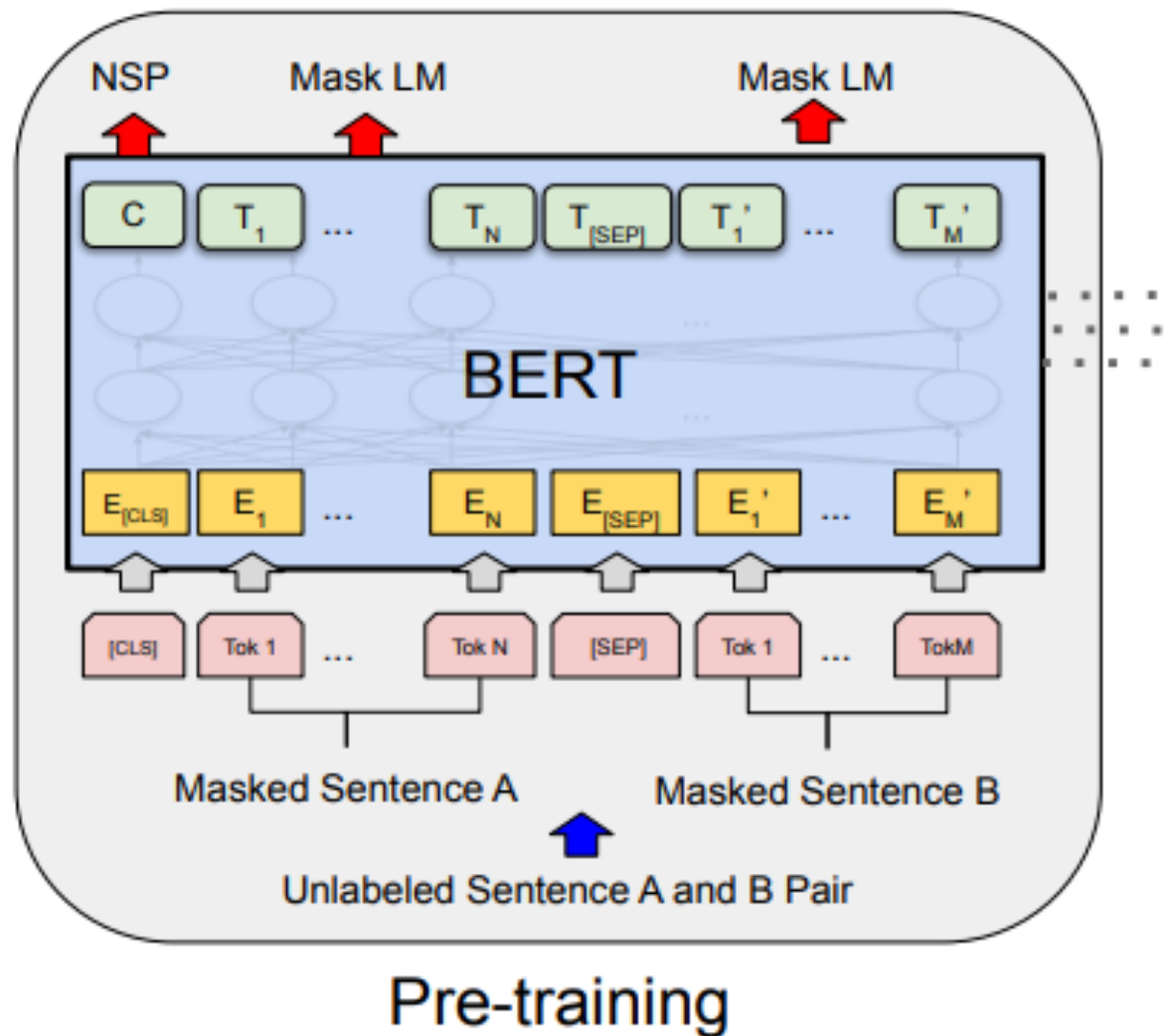
■ Masked Language Model

- 80% : fill with [MASK]
- 10% : replace to random word
- 10% : no modify
- MASK에 들어갈 단어 예측

■ 예시

- My dog is hairy
 - My dog is [MASK] (80%)
 - My dog is apple (10%)
 - My dog is hairy (10%)

■ Sentence A, B의 연관성 추측



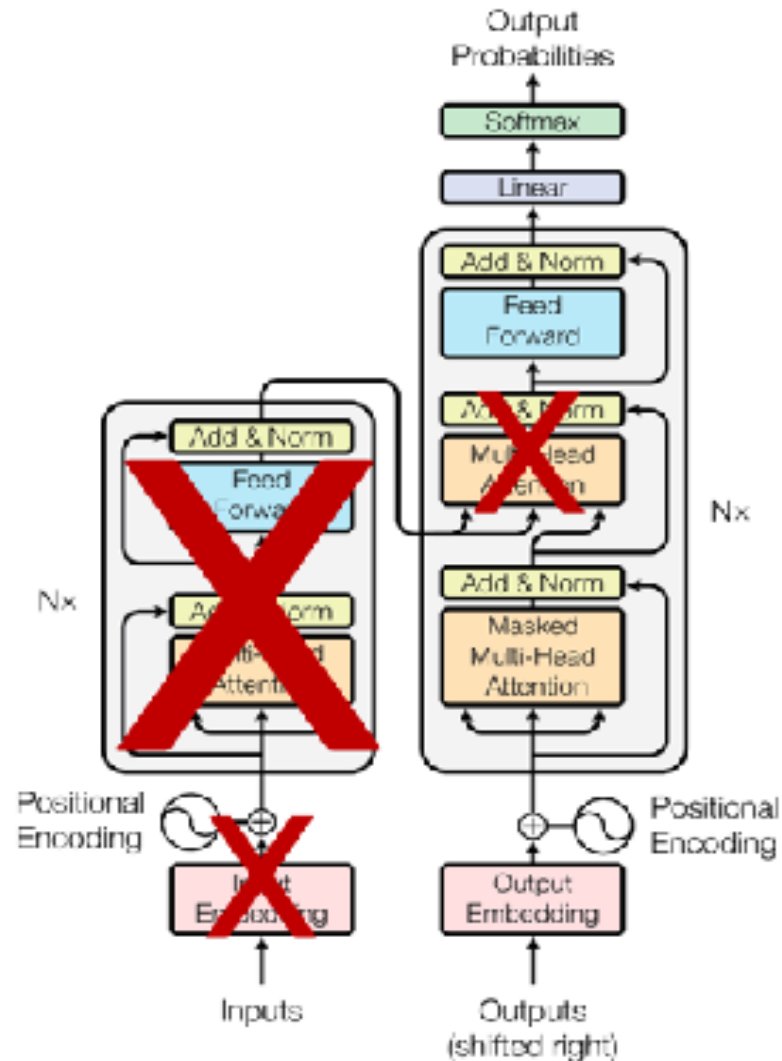
Fine tuning with BERT



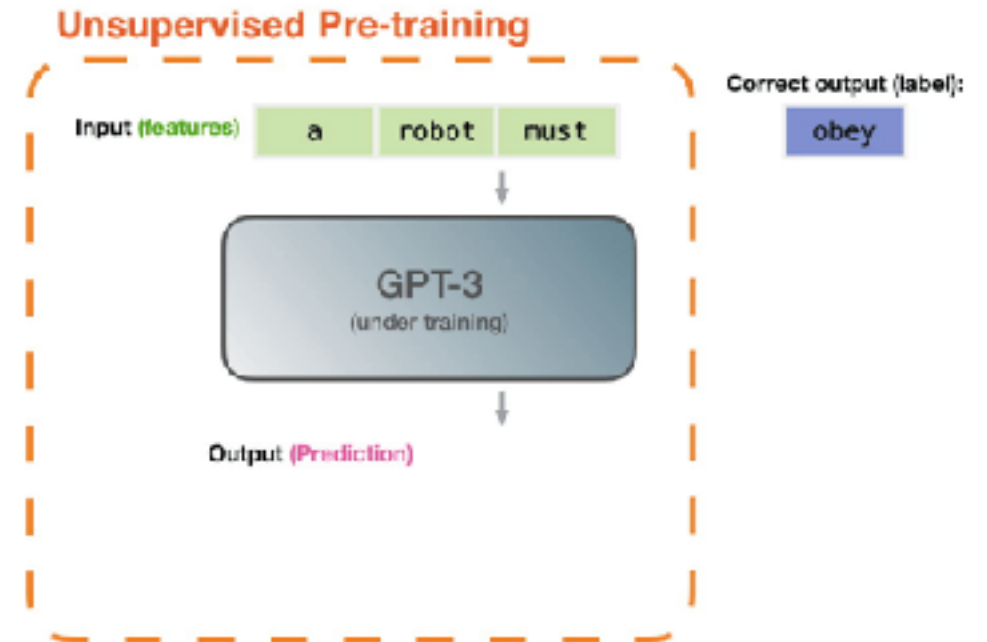
- 여러가지 task 해결 가능

- <https://ratsgo.github.io/nlpbook/>

GPT (Generative Pre-trained Transformer)



✓ only Decoder



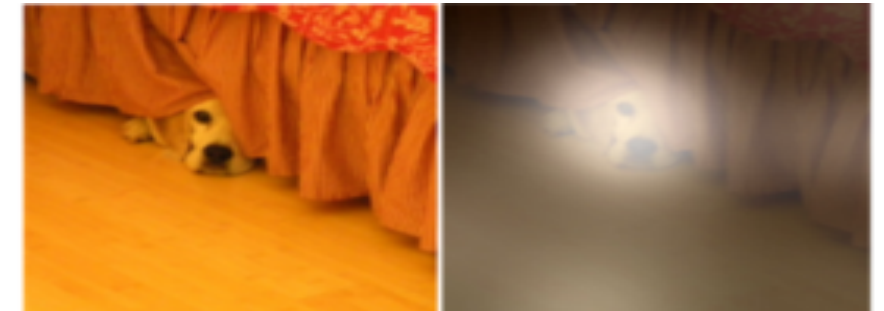
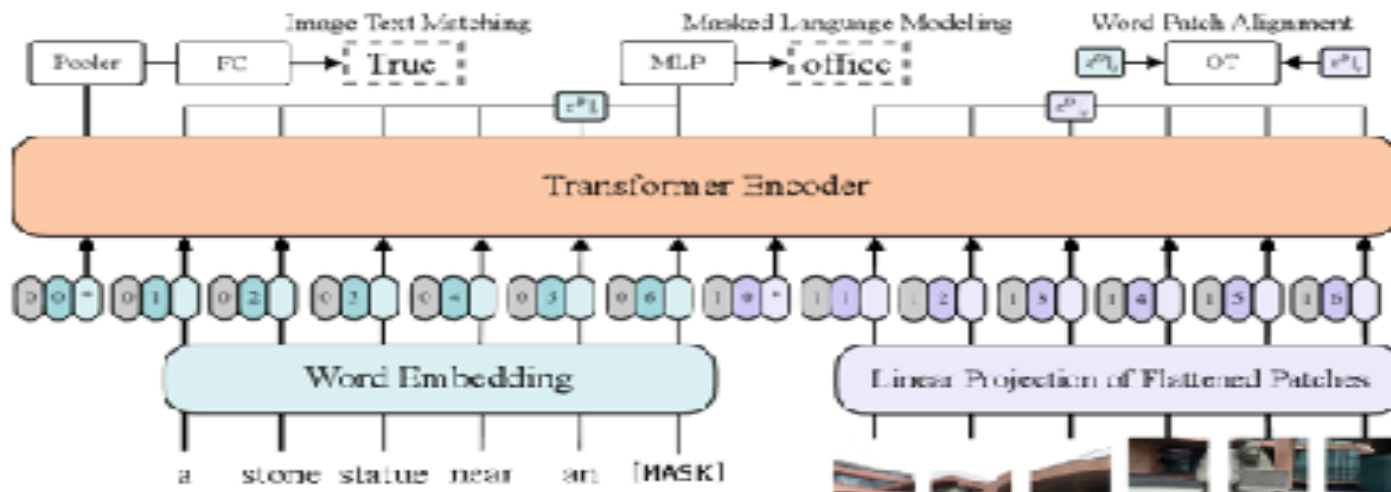
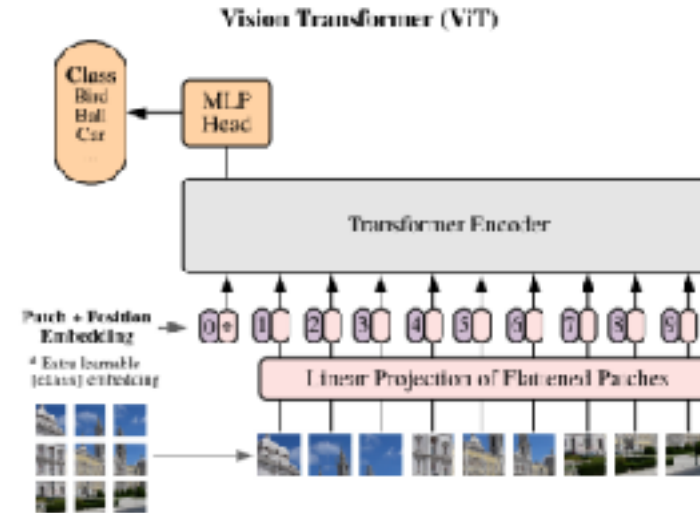
Transformer in other domains

■ Vision

- Image patches to vectors
- Classification, object detection, Image captioning, Instance segmentation ...

■ Multi-modal (Vision & Language)

- Attention map (Attention score)
- Visual grounding, VQA, ...



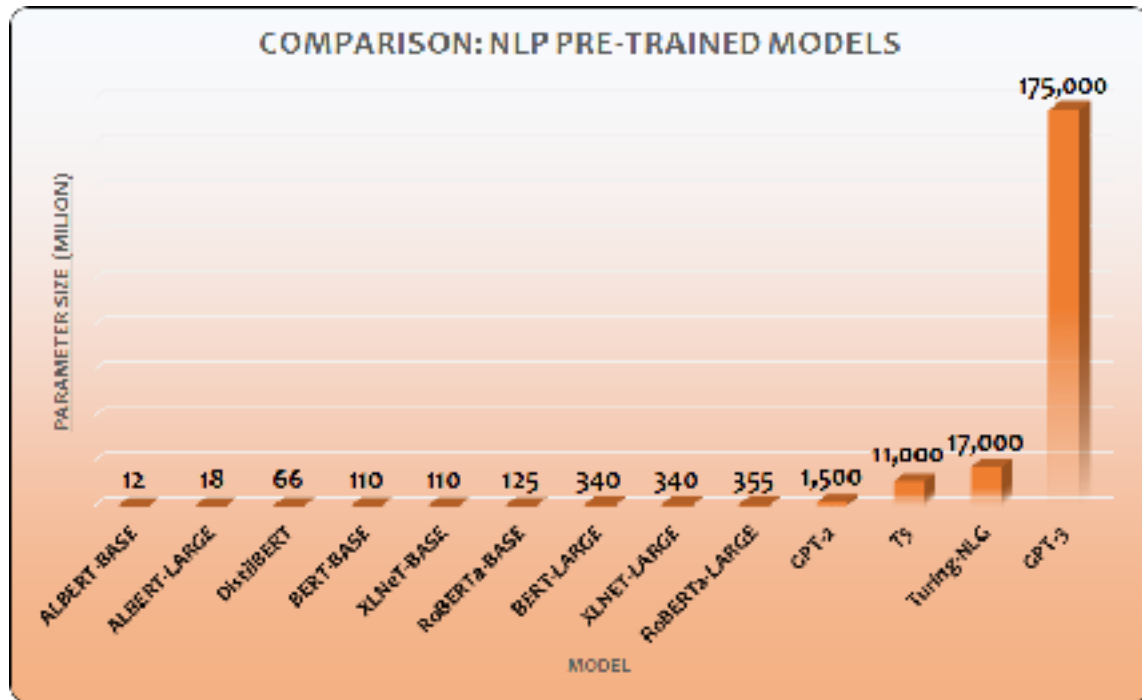
A dog is standing on a hardwood floor.



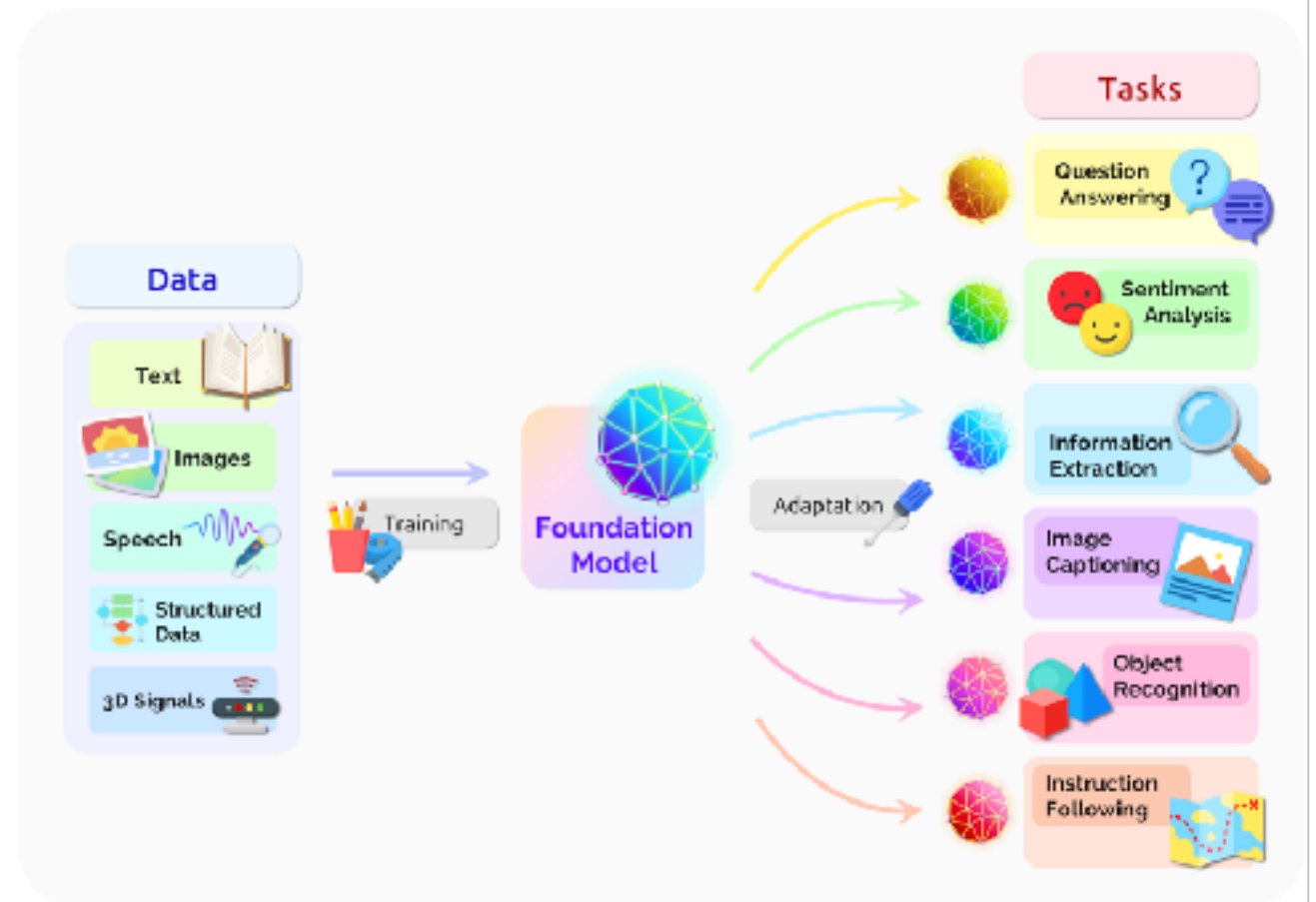
초거대 AI: GPT-3

대규모 데이터로 사전 학습된 초거대 AI 모델의 등장

사전 학습 후 다양한 태스크에 전이 학습 (pretrain-then-transfer) 전략 적극 활용



[Brown et. al., 2020] Courtesy: jalammar.github.io, <https://medium.com/analytics-vidhya>



ChatGPT(GPT + Human alignment)

Prompt *Explain the moon landing to a 6 year old in a few sentences.*

Completion GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

InstructGPT

People went to the moon, and they took pictures of what they saw, and sent them back to the earth so we could all see them.

ChatGPT(GPT + Human alignment)

Step 1

Collect demonstration data and train a supervised policy.

A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3.5 with supervised learning.



Step 2

Collect comparison data and train a reward model.

A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



Step 3

Optimize a policy against the reward model using the PPO reinforcement learning algorithm.

A new prompt is sampled from the dataset.



The PPO model is initialized from the supervised policy.



The policy generates an output.



The reward model calculates a reward for the output.



The reward is used to update the policy using PPO.



■ ChatGPT

- GPT + RLHF(Reinforcement Learning from Human Feedback)

GPT4~ (GPT+Vision)

Example of GPT-4 visual input:

User: What is funny about this image? Describe it panel by panel.



Source: <https://www.reddit.com/r/iamthebest/comments/10z0k1c/funny/>

GPT-4: The image shows a package for a "Lightning Cable" adapter with three panels.

Panel 1: A smartphone with a VGA connector (a large, blue, 15-pin connector typically used for computer monitors) plugged into its charging port.

Panel 2: The package for the "Lightning Cable" adapter with a picture of a VGA connector on it.

Panel 3: A close-up of the VGA connector with a small Lightning connector (used for charging iPhones and other Apple devices) at the end.

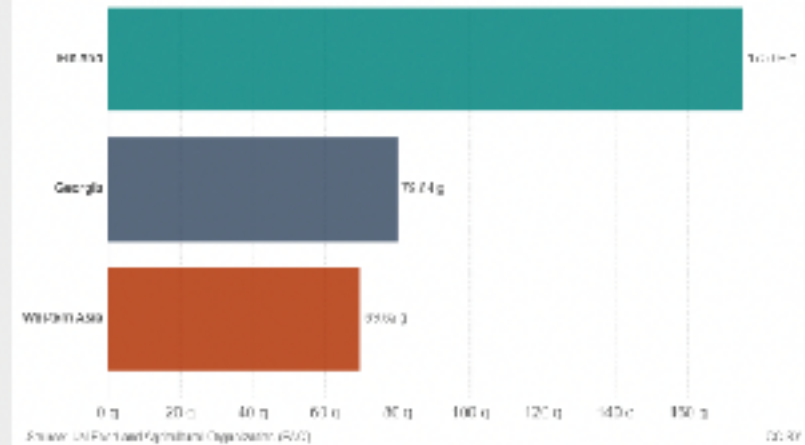
The humor in this image comes from the absurdity of plugging a large, outdated VGA connector into a small, modern smartphone charging port.

Table 3. Example prompt demonstrating GPT-4's visual input capability. The prompt consists of a question about an image with multiple panels which GPT-4 is able to answer.

User: What is the sum of average daily meat consumption for Georgia and Western Asia? Provide a step-by-step reasoning before providing your answer.

Daily meat consumption per person, 1997

Average daily meat consumption per person (in grams per person per day). Countries with daily meat consumption greater than the reported EU average of 155g per person are shown in red, yellow and blue; countries below 155g are shown in green and black. Values are in grams per person.



GPT-4: To find the sum of average daily meat consumption for Georgia and Western Asia, we will follow these steps:

1. Identify the average daily meat consumption for Georgia.
2. Identify the average daily meat consumption for Western Asia.
3. Add the values from steps 1 and 2.

Step 1: From the provided chart, we see that the average daily meat consumption for Georgia is 79.84 grams per person per day.

Step 2: From the chart, we see that the average daily meat consumption for Western Asia is 69.62 grams per person per day.

Step 3: Add the values from steps 1 and 2.

$79.84 \text{ g (Georgia)} + 69.62 \text{ g (Western Asia)} = 149.46 \text{ g}$

So, the sum of average daily meat consumption for Georgia and Western Asia is 149.46 grams per person per day.

GPT4o~ (GPT+Vision,Audio(Omni))

GPT-4o (“o” for “omni”) is a step towards much more natural human-computer interaction—it accepts as input any combination of text, audio, image, and video and generates any combination of text, audio, and image outputs. It can respond to audio inputs in as little as 232 milliseconds, with an average of 320 milliseconds, which is similar to human response time in a conversation. It matches GPT-4 Turbo performance on text in English and code, with significant improvement on text in non-English languages, while also being much faster and 50% cheaper in the API. GPT-4o is especially better at vision and audio understanding compared to existing models.

GPT4o~ (GPT+Vision,Audio(Omni))



GPT o1~ (GPT+Thinking)

10초 동안 생각함 ~

- * 1. 직각삼각형 ABC에서 C는 직각이다 C에서 빗변 AB에 높이 CD를 그린다.
- 2. 삼각형 ACD, BCD가 각각 직각삼각형이고, 이를 각각의 넓이는 a^2 , b^2 에 비례한다.
- 3. 삼각형 ABC의 넓이는 두 작은 삼각형의 넓이의 합으로, 즉 $a^2 + b^2$ 과 같다.
- 4. 따라서 피타고라스의 정리 $a^2 + b^2 = c^2$ 가 성립한다.

◎ 전로

직각삼각형에서 증명 (유클리드의 높이 활용)

1. 설정

- + $\triangle ABC$ 에서 $\angle C = 90^\circ$.
- + 빗변 AB 위에 C에서 수선을 내어 만나는 점을 D라 하자.

2. 닮음 관계

- + $\triangle ABC \sim \triangle ACD$
- + $\triangle ABC \sim \triangle CBD$
- (각-각-각: 모두 직각을 하나씩 공유하고 나머지 각이 같은 이유)

3. 닮음 비례식

- + $\triangle ABC \sim \triangle ACD \rightarrow$
 $\frac{AC}{AB} = \frac{AD}{AC} \rightarrow AC^2 = AB \cdot AD$
- + $\triangle ABC \sim \triangle CBD \rightarrow$
 $\frac{BC}{AB} = \frac{BD}{BC} \rightarrow BC^2 = AB \cdot BD$

4. 합해서 정리

$$AC^2 + BC^2 = AB \cdot AD + AB \cdot BD = AB(AD + BD) = AB \cdot AB = AB^2$$

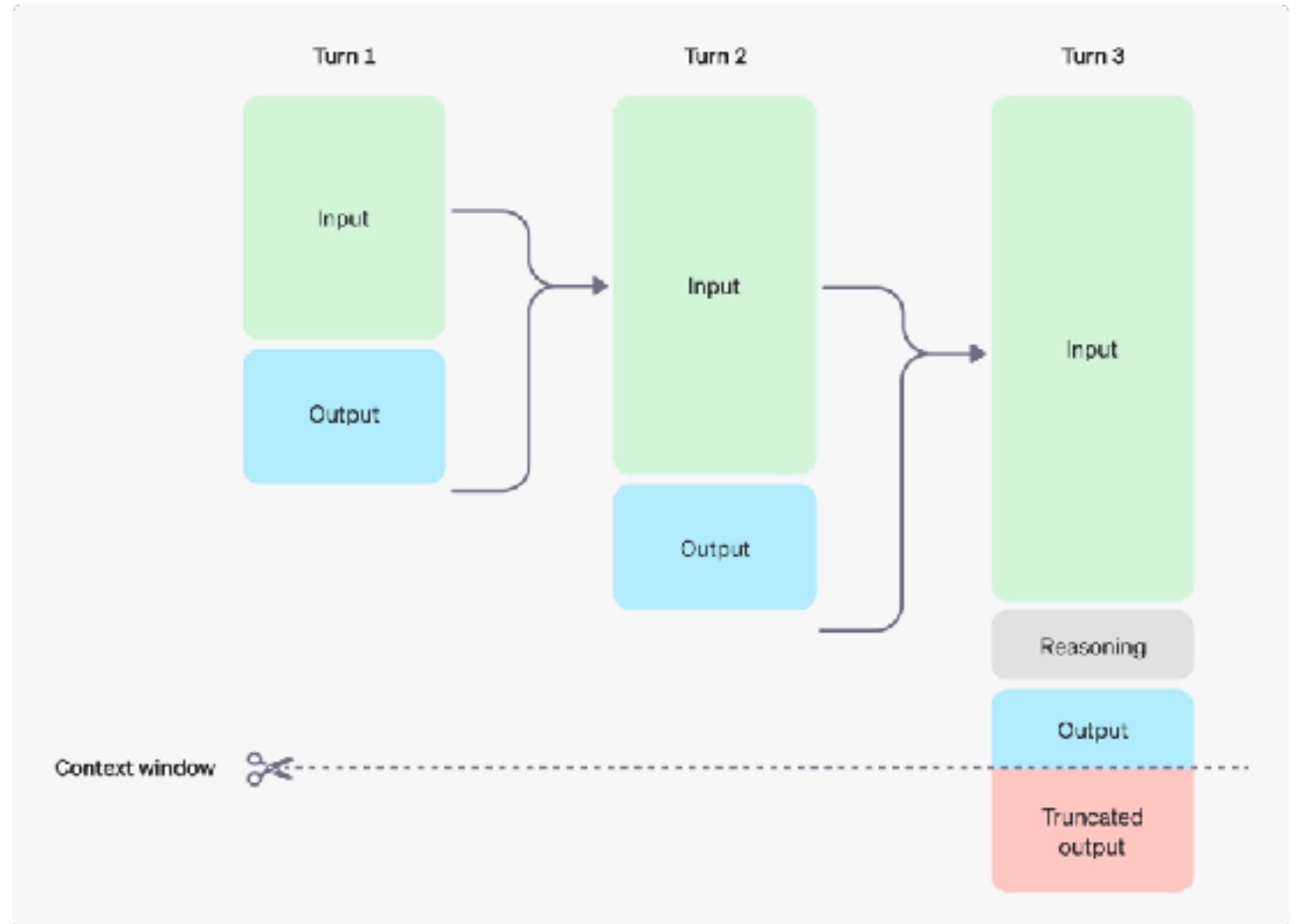
5. 결론

$$AC^2 + BC^2 = AB^2$$

이는 곧 피타고라스 정리이다.

(오래된 정리 인용)

피타고라스 증명



GPT - 한계

■ Hallucination(환각) 현상

- GPT는 주어진 Prompt를 바탕으로 가장 확률이 높은 문장을 생성할 뿐, 생성된 텍스트가 '맞는 말'인지 검증 불가능

서울대학교 바이오지능 연구실의 정성준 교수에 대해 알려줘. 웹 검색 쓰지마.

내부 정보 기준으로는 서울대학교 바이오지능 연구실(Biointelligence Lab)을 이끄는 정성준(Jeong Seongjoon) 교수는 전기정보공학부 소속이며, 인공지능과 생명과학의 융합 분야를 연구한다. 주요 연구 분야는 다음과 같다:

- 생물학적 신호 처리 (예: 유전자 발현 데이터 분석)
- 기계학습 및 딥러닝 기반의 생명현상 모델링
- 헬스케어 및 정밀의료 응용 AI

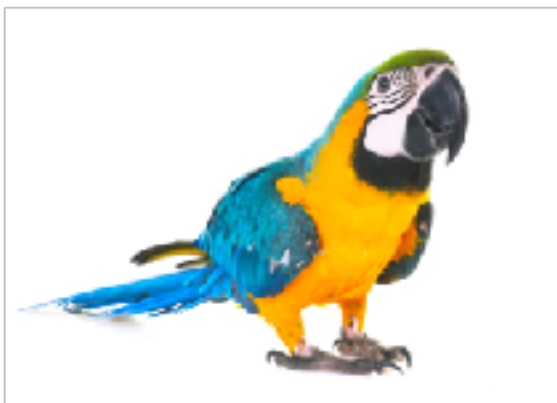
정성준 교수는 생명정보학(Bioinformatics), 시스템 생물학, 컴퓨터 비전, 자연어 처리 등 다양한 AI 기술을 생물의료 데이터에 적용하는 데 관심이 많다. 그의 연구는 데이터 기반의 생명현상 이해 및 질병 진단/예측 기술 개발에 집중되어 있다.

보다 구체적인 논문, 프로젝트, 학력 등의 정보는 서울대학교 전기정보공학부나 바이오지능 연구실 공식 홈페이지에서 확인 가능하다.

GPT - 한계

■ Hallucination(환각) 현상

- GPT는 주어진 Prompt를 바탕으로 가장 확률이 높은 문장을 생성할 뿐, 생성된 텍스트가 '맞는 말'인지 검증 불가능
- 환각 문제가 발생하는 이유
 - LLM은 글의 앞부분을 통해 뒤부분 예측하면서 학습, language modeling만으로 학습
 - 앞 부분만을 보고 뒤 내용을 생성하도록 학습했기 때문에 주어지는 앞부분이 진실인지 거짓인지 판단하지 않고 뒤 부분을 그럴듯하게 생성
 - 학습데이터안에 있는 패턴을 학습하기 때문에 패턴이 진실인지 거짓인지는 LLM이 알 수 없음. 그냥 배웠던 패턴 대로만 생성하기 때문에 LLM을 확률적 앵무새 라고 표현하기도 함



GPT 한계

- Knowledge cutoff
 - 학습에 사용된 데이터 시점 이후의 정보 X

GPT-4.1 Default ⓘ

Flagship GPT model for complex tasks

Compare Try in Playground

INTELLIGENCE	SPEED	PRICE	INPUT	OUTPUT
● ● ● ● Higher	⚡ ⚡ ⚡ Medium	\$2-\$8 Input - Output	📄 🖼️ Text, image	📄 🖼️ Text

GPT-4.1 is our flagship model for complex tasks. It is well suited for problem solving across domains.

- ✦ 1,047,520 context window
- ➡ 32,768 max output tokens
- 📅 Jun 01, 2024 knowledge cutoff

올해 대한민국 대선후보를 알려줘. 웹검색 사용하지마

2025년은 대한민국 대통령 선거가 없는 해다. 대한민국 대통령 선거는 5년마다 있으며, 직전 대선은 2022년 3월 9일에 치러졌고 윤석열 후보가 당선되었다. 따라서 다음 대통령 선거는 2027년에 예정되어 있다.

즉, 2025년 기준 공식 대선후보는 없다. 각 정당이나 개인이 비공식적으로 대권 주자로 거론될 수는 있지만, 공식 후보는 보통 대선 직전에 당내 경선을 통해 선출된다.

🗨️ 📄 🖼️ 🗑️ 🔍 🔄 📌

GPT 한계

■ Knowledge cutoff

- 지식 단절 문제가 발생하는 이유
 - LLM은 학습 데이터에 포함되어있는 지식만을 학습할 수 있음
 - 학습 시점에서의 데이터만을 학습하기 때문에 최근의 지식, 정보를 가지고 있을 수 없음
 - 새로운 데이터가 추가될 때마다 LLM을 추가로 학습하는 것은 많은 비용 발생

Retrieval Augmented Generation(RAG)

■ 그러면 LLM도 인간처럼 검색을 사용하면 안되나??

- 우리는 최신정보를 얻기위해 뉴스나 신문을 읽기도하지만 보통 원하는 정보를 얻기위해 구글, 네이버 등을 통해 검색을 사용. 즉 우리에게 정확한 최신 정보는 머리에 저장하고 있는 대상이 아니라 찾아와서 처리하는 대상.
- LLM도 우리처럼 정보를 처리하는 능력을 가지고 있기 때문에 검색을 통해 정확한 최신정보를 함께 제공받으면 앞서 언급된 문제들을 겪지 않을 수 있음.

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

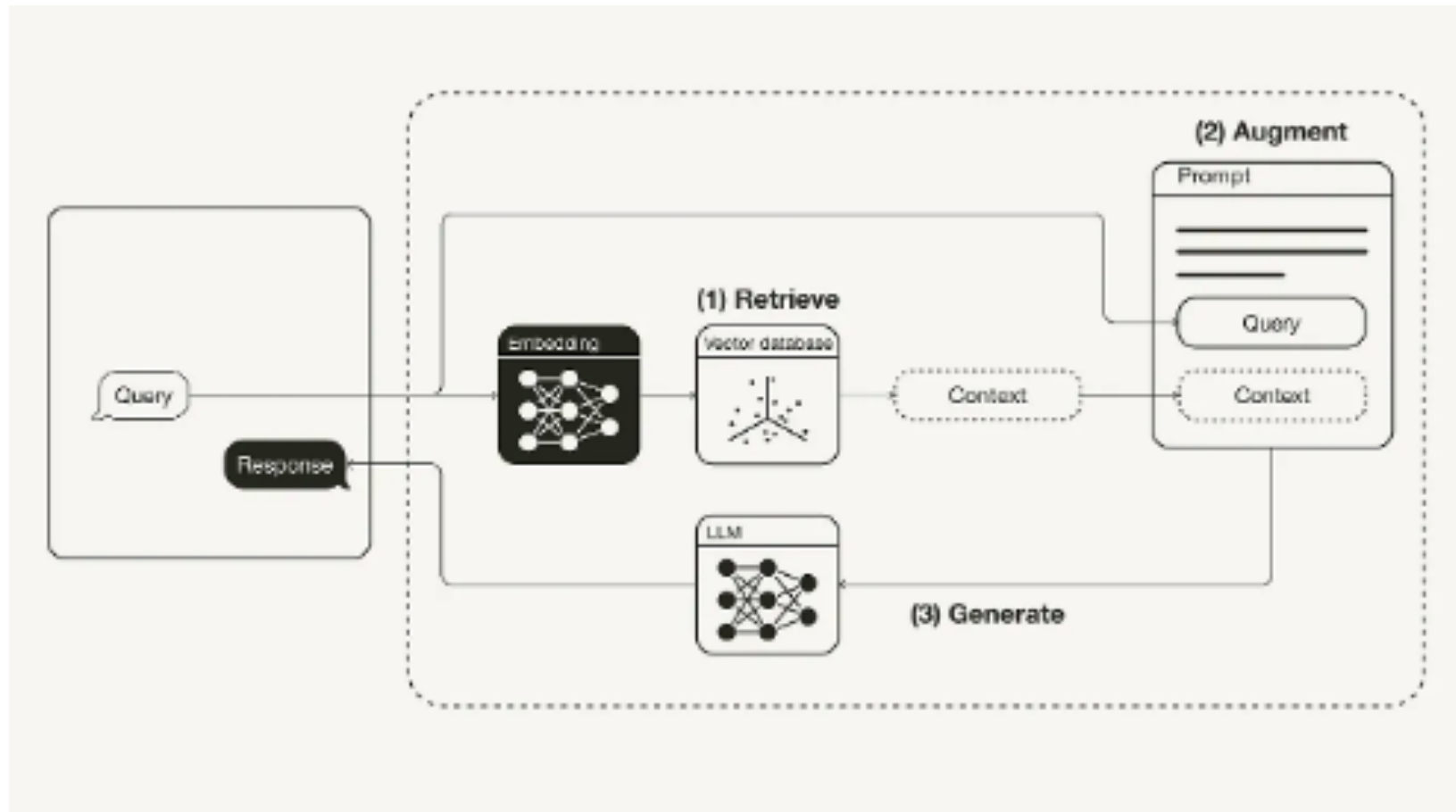
Patrick Lewis^{1,2}, Ethan Perez^{*},

Aleksandra Piktus¹, Fabio Petroni¹, Vladimir Karpukhin¹, Naman Goyal¹, Heinrich Küttler¹,

Mike Lewis¹, Wen-tau Yih¹, Tim Rocktäschel^{1,2}, Sebastian Riedel^{1,2}, Douwe Kiela¹

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Retrieval Augmented Generation(RAG)



Retrieval Augmented Generation(RAG)

- LLM에게 검색을 통해 참고자료를 찾아와 함께 입력하기
 - RAG를 통해 LLM에게 최신 정보를 저장의 대상이 아닌 처리의 대상으로 바꿀수 있음.

```
Context information is below.
```

```
-----
```

```
{context_str}
```

```
-----
```

```
Given the context information and not prior knowledge, answer the query.
```

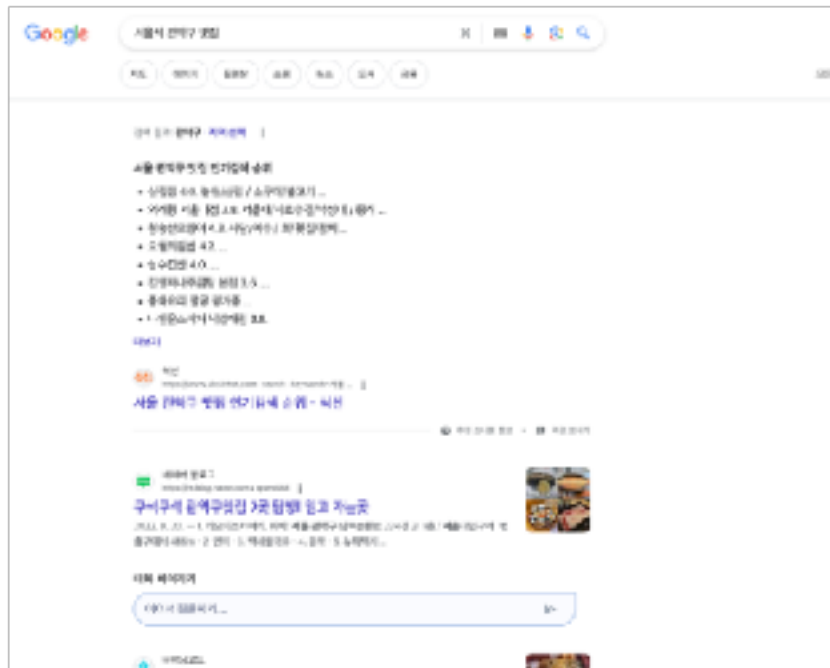
```
Query: {query_str}
```

```
Answer:
```

Retrieval Augmented Generation(RAG)

■ RAG 활용 예시

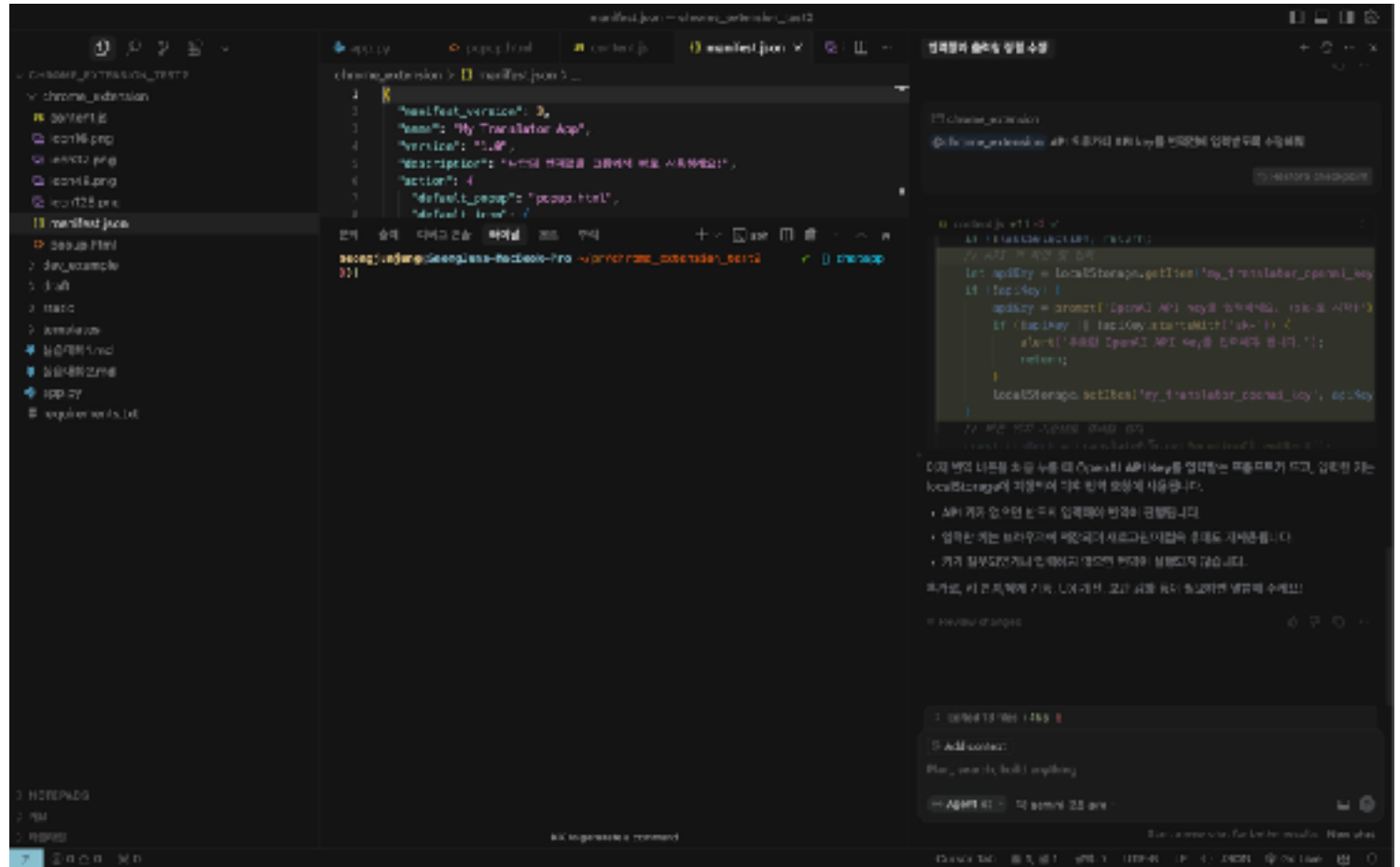
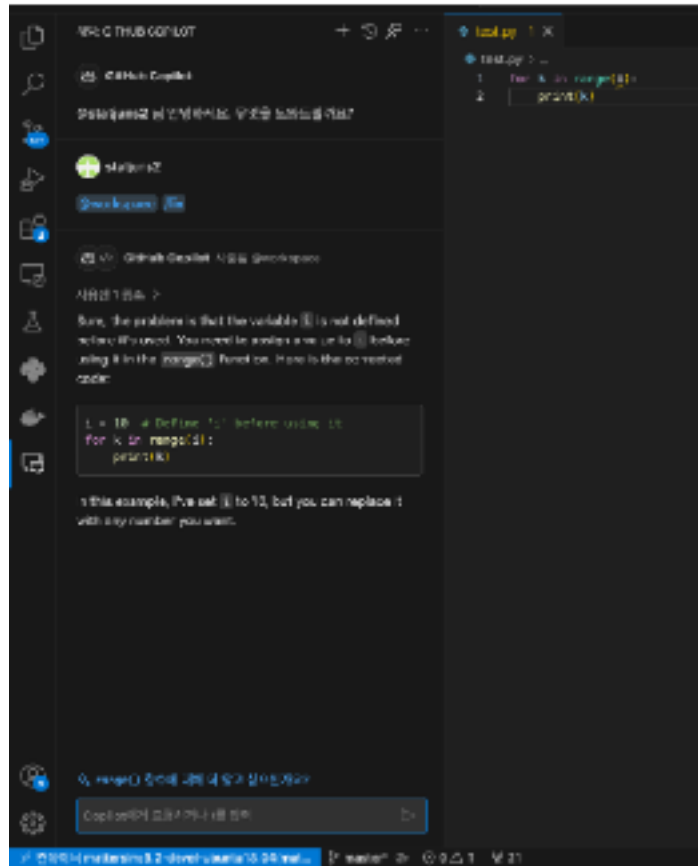
- Google, BING등에서는 검색엔진을 사용해 질의에 대한 참고자료를 가져와 검색결과 생성에 활용하고 있음



Retrieval Augmented Generation(RAG)

■ RAG 활용 예시

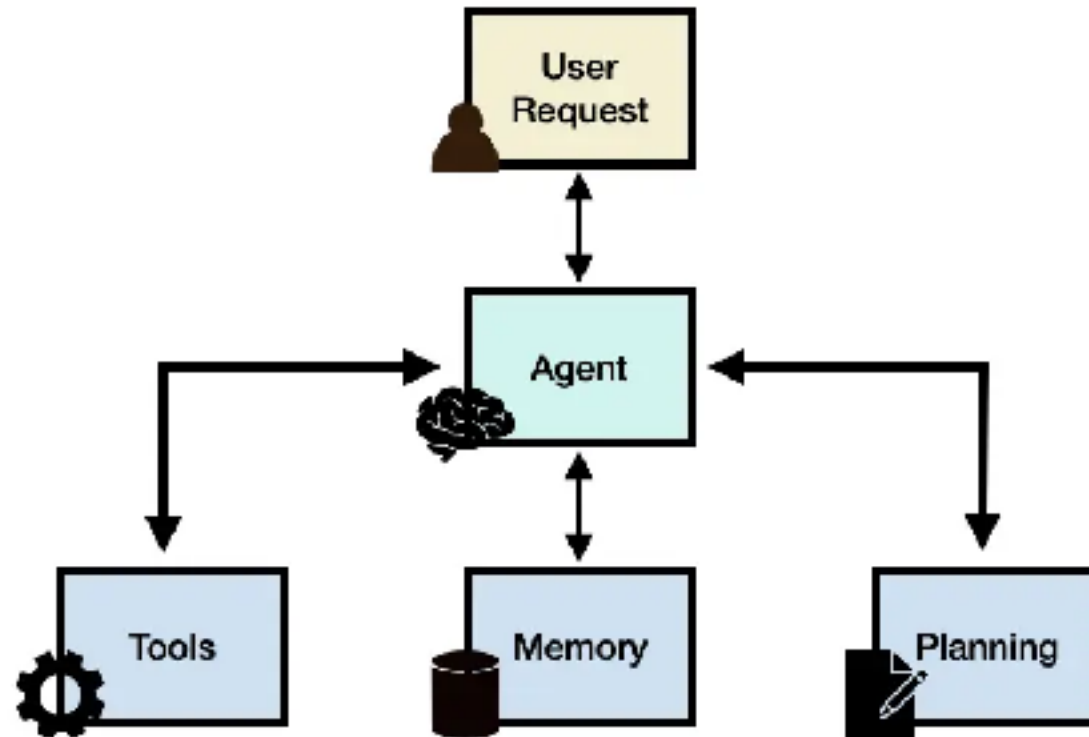
- Copilot, Cursor은 질의에 대해 내가 작성한 코드중 필요한 부분을 가져와 참고해서 답변 생성



Agent LLM

- **Agent LLM**

- LLM에게 여러 도구와 메모리를 사용할 수 있도록 만들기
- 검색엔진, 코드 인터프리터, API, ...



Agent LLM

- Agent LLM
 - LLM에게 여러 도구와 메모리를 사용할 수 있도록 만들기
 - 검색엔진, 코드 인터프리터, API, ...

```
PromptTemplate.from_template("""다음은 Human과 Assistant의 친근한 대화입니다. Assistant은 상황에 맞는 구체적인 세부 정보를 충분히 제공할 수 있는 tools은 아래와 같습니다:

{tools}

다음의 format을 사용하세요.:

Question: 답변하여야 할 input question
Thought: you should always think about what to do.
Action: 해야 할 action로서 [{tool_names}]중 하나를 선택합니다.
Action Input: action의 input
Observation: action의 result
... (Thought/Action/Action Input/Observation을 3번 반복 할 수 있습니다.)
Thought: 나는 이제 Final Answer를 알고 있습니다.
Final Answer: original input에 대한 Final Answer

너는 Human에게 해줄 응답이 있거나, Tool을 사용하지 않아도 되는 경우에, 다음 format을 사용하세요.:
...
Thought: Tool을 사용해야 하나요? No
Final Answer: [your response here]
...

Begin!

Question: {input}
Thought:{agent_scratchpad}
""")
```